Case Study

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nudge

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ME 314 Advanced Product Development

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Contact / Bio



Hi, I'm Tom.

Ever since I can remember, I thrived to become a "rocket scientist". Now thriving to impact and excite the world by creating innovative solutions to current real-life problems.

My background has taken me across continents. I was born and raised in Korea, spent four years in Brazil during elementary school, and after graduating high school, flew to New York to study mechanical engineering at The Cooper Union.



Contact Me!

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Problem Scope

Food waste is caused because food disappears when it is out of sight.



30 – 40 %

of U.S. food goes uneaten

(United States Environmental Protection Agency)

83 %

Reported spoiled or stale as most common cause of food waste

(International Food Information Council)

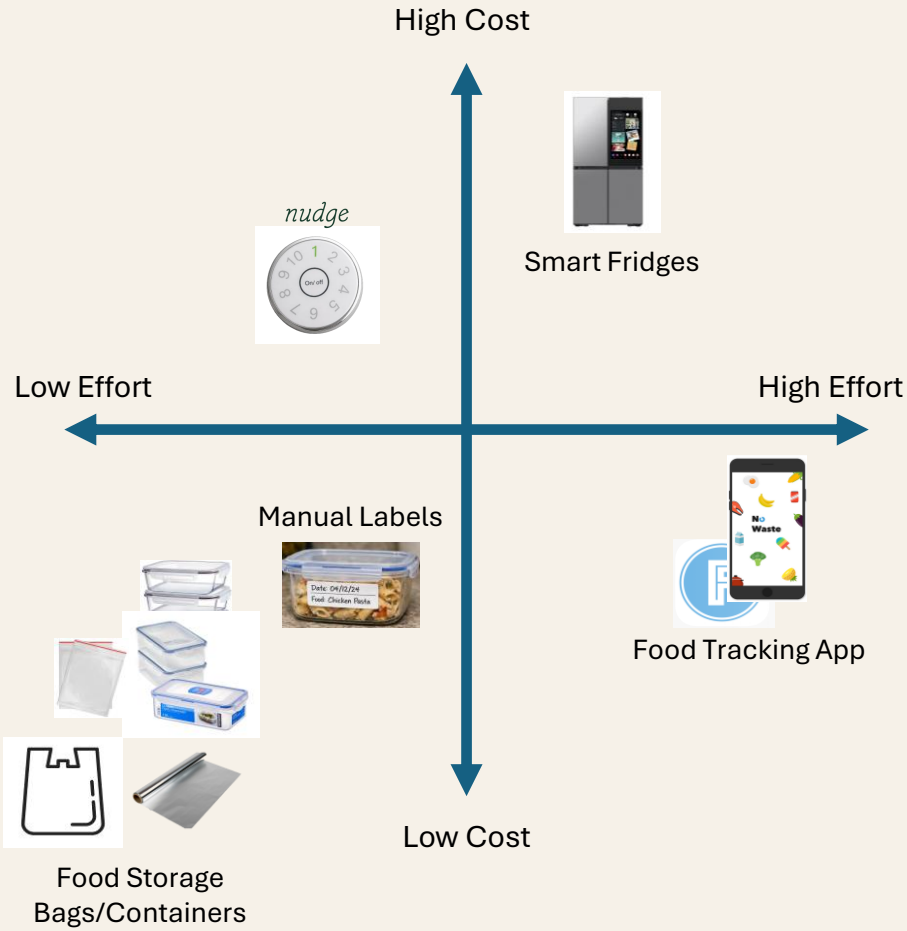
1 in 3

People forget leftovers once they are out of sight

(New York Post)

Market Research

Existing food reminder solutions are either expensive, effort-heavy, or not active reminders.



Market Gap

There is a clear gap between high tech solutions (smart fridges, apps) and Low-tech solutions (plastic containers)

Primary User Group

Busy students, young professionals, and shared-kitchen users who store leftovers but experience food waste due to limited fridge space and poor food visibility.



Persona

- Name: Joshua Lee
- Age: 24
- Location: Manhattan, NY
- Living Situation: Shares an apartment with a roommate
- Occupation: Entry Software Engineer

Problems they Face

- “My roommate pushed my leftover pizza behind his food”
- “I forgot that leftover pasta”
- “I feel bad throwing my food away”

Solution



Green – Fresh
Day 1- 3



Yellow – Consume Soon
Day 4- 7



Red – Spoiled
Day 8 - 10

nudge makes stored food visible at the moment of decision.

One-touch activation

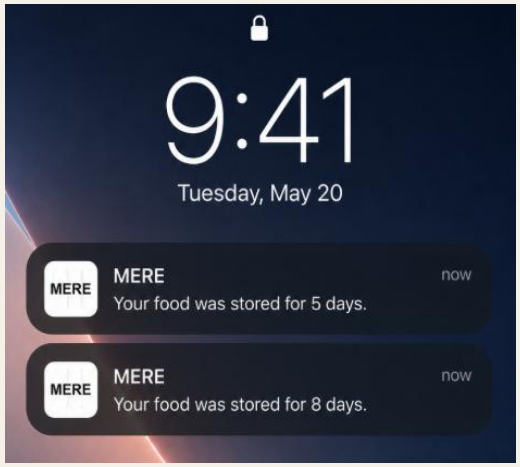
- User presses the center button to start tracking
- No setup, no app input, no configuration required

Time-based freshness indicator

- Automatically tracks how long food has been stored
- Visual LED feedback communicates status instantly

Solution





Light cues when the fridge opens

Smart notifications

- Alerts sent at critical decision points:
  Day 5 → "Consume soon"
  Day 8 → "Spoiled"
- Reduces over-notification while maximizing impact

Solution

Modular accessory ecosystem with built-in tag slot for easy attachment and removal



Silicone Sleeves



Glass Containers

Solution

Modular accessory ecosystem with built-in tag slot for easy attachment and removal

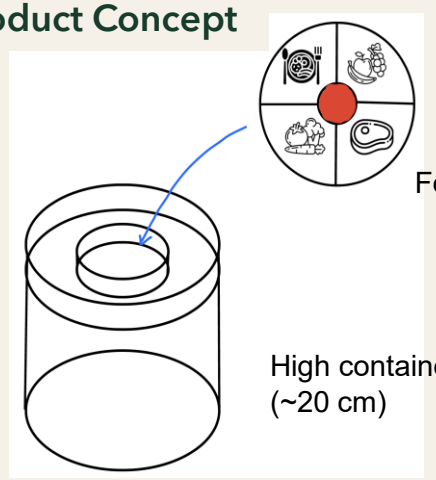


Wine Bottle Stoppers

Early Design & Ideation



Early Product Concept



Freshness Indicator on Center

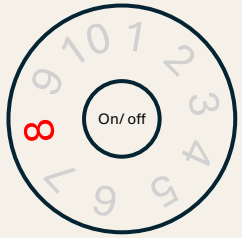
Food Type Selection

High container to avoid stacking (~20 cm)

Different Face Design







7	8	9
4	5	6
1	2	3
0		On / Off

7	8	9
4	5	6
1	2	3
0		On / Off

7	8	9
4	5	6
1	2	3
0		On / Off

Final Design Specification

Circular Day Interface



Materials

- CNC machined and anodized aluminum side rim for a premium durable finish
- Frosted polycarbonate top lens to softly diffuse LED light
- Injection molded plastic base.

Form Factor

- Ø 33.8 mm compact circular tag
- 6 mm thin profile for container and bag attachment
- Rounded edge for a soft, premium feel

Final Prototype Model





Branding

MERE

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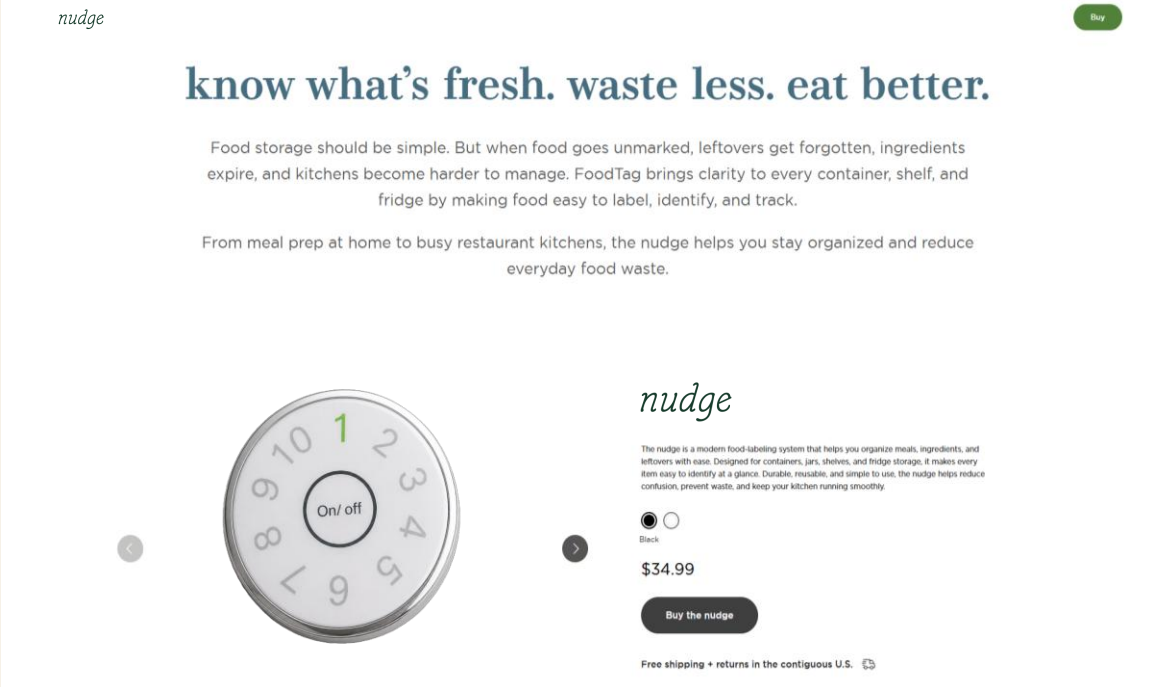
Freshness, remembered

Brand Name
MERE
Font: Avenir Next LT Pro Demi
Color: #173C2C
Letter spacing: expanded

Product Name
nudge
Font: Fraunces 9pt SuperSoft Thin
Color: #173C2C

Tagline
Freshness, remembered
Font: Avenir Next LT Pro Light
Color: #173C2C

Sales Channel

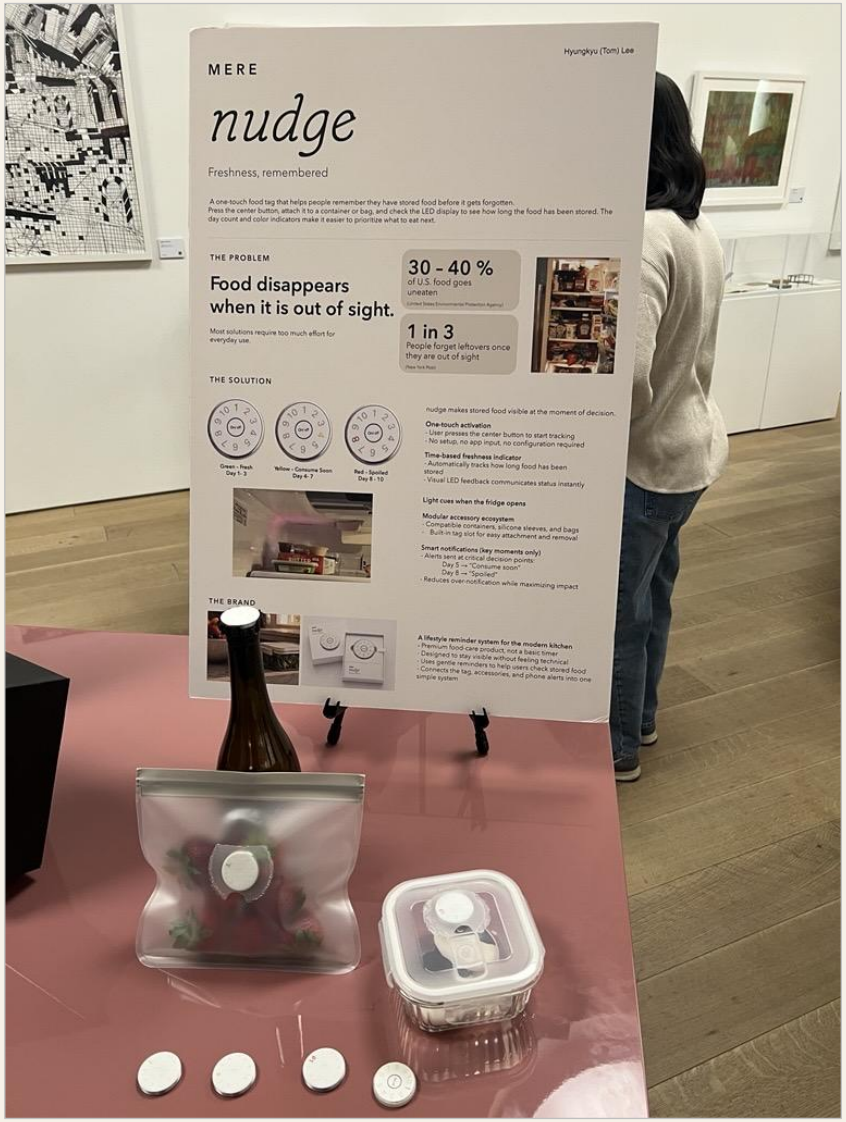


SALES CHANNEL

Direct-to-consumer online launch

- Website communicates the brand story, product purpose, and calm feeling of the tag
- Available as a single tag or bundled with containers and silicone bags
- Website explains how nudge helps users remember stored food
- Future expansion: kitchenware stores and grocery shops

Pitch Poster



MERE

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Freshness, remembered

Hyungkyu (Tom) Lee

A one-touch food tag that helps people remember they have stored food before it gets forgotten.
Press the center button, attach it to a container or bag, and check the LED display to see how long the food has been stored. The day count and color indicators make it easier to prioritize what to eat next.

THE PROBLEM

Food disappears when it is out of sight.

Most solutions require too much effort for everyday use.

30 – 40 %
of U.S. food goes uneaten
(United States Environmental Protection Agency)

1 in 3
People forget leftovers once they are out of sight
(New York Post)

THE SOLUTION

Green - Fresh Day 1- 3

Yellow - Consume Soon Day 4- 7

Red - Spoiled Day 8 - 10

nudge makes stored food visible at the moment of decision.

One-touch activation
- User presses the center button to start tracking
- No setup, no app input, no configuration required

Time-based freshness indicator
- Automatically tracks how long food has been stored
- Visual LED feedback communicates status instantly

Light cues when the fridge opens

Modular accessory ecosystem
- Compatible containers, silicone sleeves, and bags
- Built-in tag slot for easy attachment and removal

Smart notifications (key moments only)
- Alerts sent at critical decision points:
Day 5 → "Consume soon"
Day 8 → "Spoiled"
- Reduces over-notification while maximizing impact

THE BRAND

A lifestyle reminder system for the modern kitchen
- Premium food-care product, not a basic timer
- Designed to stay visible without feeling technical
- Uses gentle reminders to help users check stored food
- Connects the tag, accessories, and phone alerts into one simple system